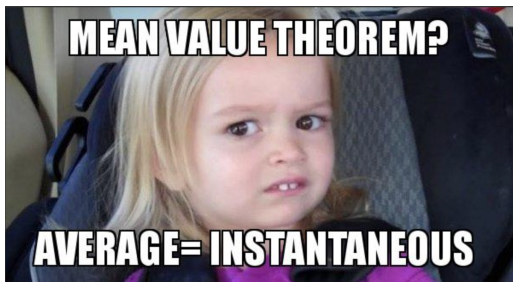


Tutorial - Week 6

TG 11 - TG 12

1. Give examples of continuous functions that satisfy the following requirements:
 - (a) $f: (a, b] \rightarrow \mathbb{R}$ bounded above but with no absolute maximum.
 - (b) $f: (a, b) \rightarrow \mathbb{R}$ with absolute minimum reached at 3 points.
 - (c) $f: [a, +\infty) \rightarrow \mathbb{R}$ with no absolute maximum and no absolute minimum.
2. Show examples of real functions on a closed interval $[a, b]$ with
 - (a) a local maximum but no global maximum.
 - (b) a global maximum reached at a point of discontinuity.
 - (c) a negative local maximum and a positive local minimum.
 - (d) infinitely many local minima.
 - (e) a global maximum that is also a local minimum, but no global minimum.
3. Find the critical numbers and classify them. Find the absolute maximum and minimum whenever they exist.
 - (a) $f(x) = |x + 1| - |x - 1|$, on $[-4, 2]$.
 - (b) $f(x) = x + \sin x$, on $[0, 4\pi]$
 - (c) $f(x) = \cos x + \sin x$, on $[0, 2\pi]$.
 - (d) $f(x) = \sqrt{x} - \sqrt[3]{x}$, on $[0, 5]$.
 - (e) $f(x) = \frac{x^2-1}{x-1}$, on $\mathbb{R}$.
 - (f) $f(x) = x^2e^{-2x}$, on $\mathbb{R}$.
4.
 - (a) Show that the continuity condition in the hypothesis of the Mean Value Theorem is necessary.
 - (b) Similarly, show that differentiability is also necessary for the Mean Value Theorem.
 - (c) If a function has a discontinuity, is it possible to have $f'(c)(b - a) = f(b) - f(a)$ for some $c \in (a, b)$?



5. Determine the intervals (if any) where the Mean Value Theorem applies.
 - (a) $\frac{1}{x^3}$.
 - (b) $\ln(5x - 3)$.
 - (c) $\sqrt{x^2 + 2}$.
 - (d) $|\sin(x) + \frac{1}{2}|$.

6. Let $f: [0, 2] \rightarrow \mathbb{R}$ be a continuous function such that

- f and f' are continuous in $[0, 2]$ and differentiable in $(0, 2)$,
- $f(0) = 0$, $f(1) = 1$, $f(2) = 4$.

Prove that there exists $c \in (0, 2)$ such that $f'(c) = d$ for $d = 1.5$, 2 , 2.5 .

7. Show that the equation $x^3 + e^x = 0$ has exactly one solution.

8. Consider the function $f(x) = \frac{2\sin(9-x^2)}{9-3x}$.

- Analyze the existence of asymptotes of f .
- Prove that

$$\tilde{f}(x) = \begin{cases} f(x) & \text{if } x \neq 3 \\ 4 & \text{if } x = 3 \end{cases}$$

is differentiable at $x = 3$.

9. Consider the functions $f(x) = x\sqrt{6-x}$ and $g(x) = \frac{2x}{3x^2+6x+3}$. Find the following for each one.

- Domain and symmetries.
- Asymptotes.
- Critical points.
- Local maximum and minimum.
- Increasing/decreasing intervals.
- Inflection points.
- Concavity intervals.
- Sketch the graph.

10. Find the limits.

- $\lim_{x \rightarrow \infty} \frac{2^x}{x^2 + 2x + 1}$.
- $\lim_{x \rightarrow 0^+} \left(\frac{1}{x^2} - \frac{1}{\tan x} \right)$.
- $\lim_{x \rightarrow 0^+} x^{\sin x}$.
- $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$.
- $\lim_{x \rightarrow \infty} e^x - x$.
- $\lim_{x \rightarrow 0^+} (1 + \sin 3x)^{\frac{1}{x}}$.
- $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}}$.
- $\lim_{x \rightarrow \infty} \frac{3x^3 - \sin x}{x^3 + 1}$.

